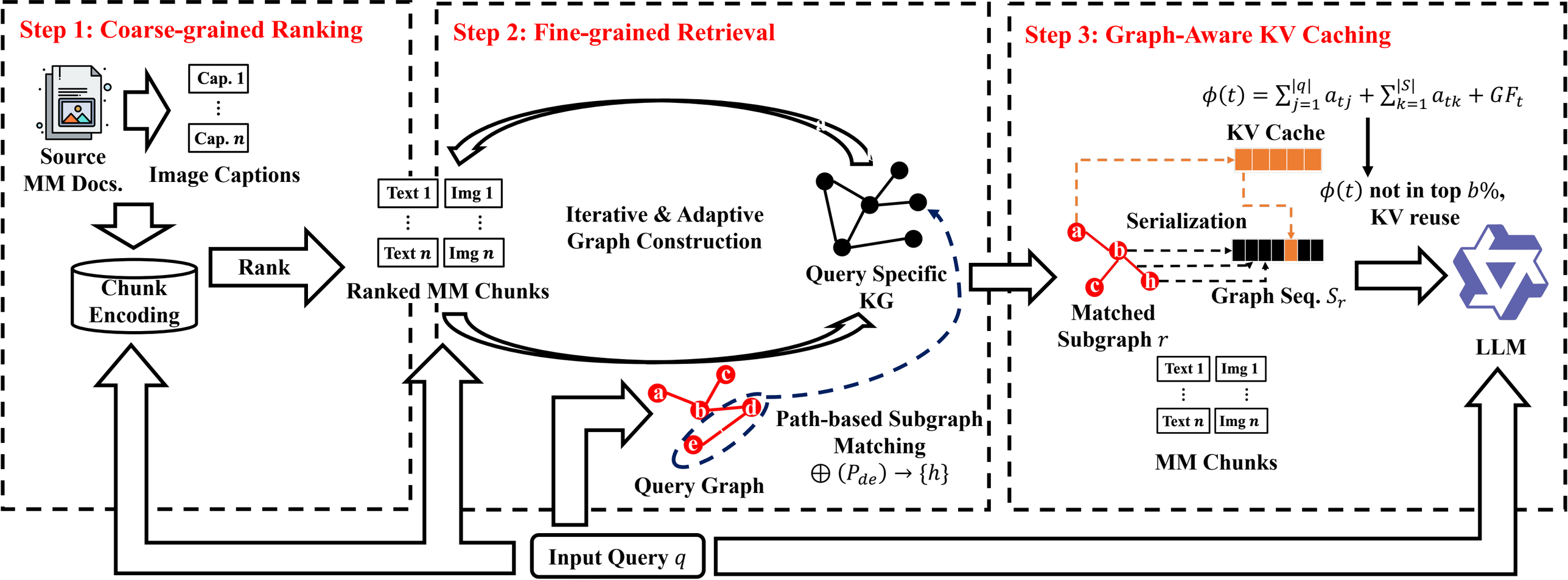

Step 1: Coarse-grained Ranking
Source MM Docs.
Cap. 1
Cap. n
Image Captions
Chunk Encoding
Rank
Step 2: Fine-grained Retrieval
Text 1
Img 1
Text n
Img n
Ranked MM Chunks
Iterative & Adaptive Graph Construction
Query Specific KG
a
b
c
d
e
Query Graph
Path-based Subgraph Matching
$\oplus (P_{de}) \rightarrow \{h\}$
Step 3: Graph-Aware KV Caching
$\phi(t) = \sum_{j=1}^{|q|} a_{tj} + \sum_{k=1}^{|S|} a_{tk} + GF_t$
KV Cache
$\phi(t)$ not in top b%, KV reuse
Serialization
Matched Subgraph r
Graph Seq. S_r
LLM
Text 1
Img 1
Text n
Img n
MM Chunks
Input Query q